# GLOBAL SUPERSTORE CLEANING AND ANALYSIS REPORT

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# SUMMARY

# INTRODUCTION

This project analyses the Global Superstore dataset from Kaggle in order to understand customers behaviour and product performance. The dataset contains sales transactions across the globe, including information about customers, products, sales, and profit.

The objective is to identify key insights such as the most valuable customers, the most profitable products, and how sales and profit vary across regions and time.

The data was cleaned and analyzed using Python and several visualizations were created to support the findings.

# DATASET OVERVIEW

This dataset covers a period from January 1, 2011 to December 31, 2014 and contains information related to:

- Order ID
- Order Date
- Ship Date
- Ship Mode
- Customer ID
- Customer Name
- Segment
- City
- State
- Country
- Market
- Region
- Product ID
- Category
- Sub-Category
- Product Name
- Sales
- Quantity
- Discount
- Profit
- Shipping Cost
- Order Priority

It has a total of 51290 rows and 24 columns.

# OBJECTIVES

- ***Customers Analysis***

1) Profile the customers based on their frequency of purchase: calculate frequency of purchase for each customer
2) Do the high frequent customers contribute more revenue?
3) Are they also profitable? What is the profit margin across the buckets?
4) Which customer segment is most profitable in each year?
5) How are the customers distributed across the countries?

- ***Product Analysis***

1) Which country has top sales?
2) Which are the top 5 profit-making product types on a yearly basis?
3) How is the product price varying with sales? Is there any increase in sales with the decrease in price at a day level?
4) What is the average delivery time across the counties?

# CLEANING

The  dataset was imported into Google Colab to clean the data using the Python programming language. Since this database was available on Kaggle, it was imported directly from that platform.

```
import pandas as pd

import kagglehub
path =
kagglehub.dataset_download("apoorvaappz/global-super-store-dat
aset")

import os
print(os.listdir(path))

import glob
files = glob.glob(path + "/*.csv")
```

```
df =
pd.read_csv("/kaggle/input/global-super-store-dataset/Global_S
uperstore2.csv", encoding='latin1')
df.head(30)
```

```
list(df.columns)
```

```
#Check for duplicates
df.duplicated().sum()
```

```
#Check for missing values
df.isnull().sum()
```

Postal Code has 41296 missing values, which represents approximately 80% of the dataset. Since this data can’t change our analysis, we decided to drop it.

```
#Delete Postal Code column
df = df.drop(columns=['Postal Code'])
```

```python
#Convert datetime
df['Order Date'] = pd.to_datetime(df['Order Date'],
dayfirst=True)
df['Ship Date'] = pd.to_datetime(df['Ship Date'],
dayfirst=True)
```

```python
df['Order Date'].dtype
-> dtype('<M8[ns]')
```

```python
#Format / type errors
df.dtypes
```

```python
#Check input errors
cols = ['Ship Mode', 'Segment', 'City','State','Country',
'Market', 'Region','Category', 'Sub-Category',
       'Product Name','Order Priority']

for col in cols:
  print(col)
  print(df[col].unique())
```

```python
#Check for inconsistent values
df.describe()
```

# EXPLANATORY ANALYSIS

- **Customers Analysis**

*Question 1: “Profile the customers based on their frequency of purchase: calculate frequency of purchase for each customer”*

```
#Customer purchase count
purchase_count = df['Customer ID'].value_counts()
purchase_count
```

| | count |
|---|---|
| **Customer ID** | |
| **PO-18850** | 97 |
| **BE-11335** | 94 |
| **JG-15805** | 90 |
| **SW-20755** | 89 |
| **MY-18295** | 85 |
| **...** | ... |
| **BG-1035** | 1 |
| **MG-7650** | 1 |
| **ME-8010** | 1 |
| **MG-7890** | 1 |
| **ZC-11910** | 1 |

```
purchase_count.describe()
```

| | count |
|---|---|
| **count** | 1590.000000 |
| **mean** | 32.257862 |
| **std** | 21.910515 |
| **min** | 1.000000 |
| **25%** | 12.000000 |
| **50%** | 28.000000 |
| **75%** | 52.000000 |
| **max** | 97.000000 |

```python
#Bucketing purchase frequencies
def get_bucket(counts):
  if counts <= 12:
    return "Low"
  elif counts <= 52:
    return "Medium"
  else:
    return "High"
```

```python
#Mapping get_bucket to purchase_count
buckets = purchase_count.map(get_bucket)
buckets
```

| | count |
|---|---|
| **Customer ID** | |
| **PO-18850** | High |
| **BE-11335** | High |
| **JG-15805** | High |
| **SW-20755** | High |
| **MY-18295** | High |

| | |
|---|---|
| ... | ... |
| **BG-1035** | Low |
| **MG-7650** | Low |
| **ME-8010** | Low |
| **MG-7890** | Low |
| **ZC-11910** | Low |

```
#Add buckets to the df
df['Purchase Frequency'] = df['Customer ID'].map(buckets)
```

```
buckets.value_counts()
```

| | **count** |
|---|---|
| **count** | |
| **Medium** | 783 |
| **Low** | 431 |
| **High** | 376 |

*Question 2: "Do the high frequent customers contribute more revenue?"*

```
#Customers generating the most revenue
sales = df.groupby('Purchase Frequency')['Sales'].sum()
sales
```

| Purchase Frequency | Sales |
| --- | --- |
| High | 6.268688e+06 |
| Low | 5.834987e+05 |
| Medium | 5.790315e+06 |

**The analysis shows that high frequency customers generate the highest revenue contribution. This suggests a positive relationship between purchase frequency and sales generation.**

*Question 3:* “Are they also profitable? What is the profit margin across the buckets?”

```
#Customers generating the most profit
profit_margin = df.groupby('Purchase
Frequency')['Profit'].sum()
profit_margin
```

| Purchase Frequency | Profit |
| --- | --- |
| High | 755084.55418 |
| Low | 60769.51900 |
| Medium | 651603.21810 |

```
#Profit margin percentage
margin_percentage = (profit_margin / sales)*100
margin_percentage
```

| | 0 |
|---|---|
| Purchase Frequency | |
| High | 12.045336 |
| Low | 10.414680 |
| Medium | 11.253329 |

**The analysis shows that high frequency customers are also profitable, with the highest profit margin (~12%) across the customer frequency buckets. However, the differences between the buckets remain relatively small, suggesting that all customer groups contribute positively to profitability, although high-frequency customers perform slightly better.**

*Question 4: "Which customer segment is most profitable in each year?"*

```
#Extract year
df['year'] = df['Order Date'].dt.year
```

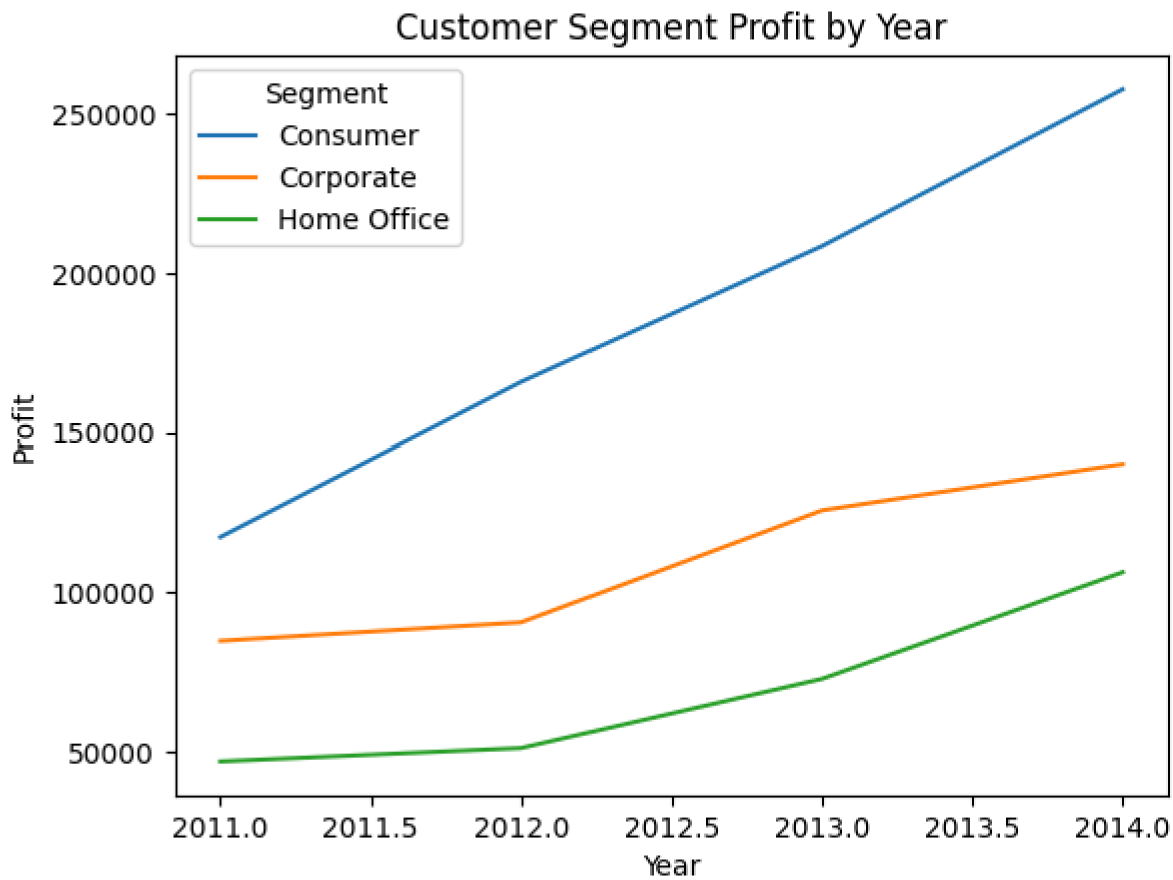
```
#Customer segment profit by year
segment_profit =
df.groupby(['year','Segment'])['Profit'].sum().reset_index()
segment_profit
```

| | year | Segment | Profit |
|---|---|---|---|
| 0 | 2011 | Consumer | 117337.49406 |
| 1 | 2011 | Corporate | 84746.93574 |
| 2 | 2011 | Home Office | 46856.38174 |
| 3 | 2012 | Consumer | 165799.19094 |
| 4 | 2012 | Corporate | 90556.69992 |
| 5 | 2012 | Home Office | 51059.38824 |

| | | | |
|---|---|---|---|
| **6** | 2013 | Consumer | 208427.73398 |
| **7** | 2013 | Corporate | 125707.93908 |
| **8** | 2013 | Home Office | 72799.55712 |
| **9** | 2014 | Consumer | 257675.36308 |
| **10** | 2014 | Corporate | 140196.75392 |
| **11** | 2014 | Home Office | 106293.85346 |

```
#Visualization
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

sns.lineplot(data = segment_profit, x= 'year', y= 'Profit',
hue= 'Segment')
plt.title("Customer Segment Profit by Year")
plt.xlabel('Year')
plt.show()
```




**The Consumer segment consistently generates the highest profit across the years, indicating a stronger contribution to overall profitability compared to Corporate and Home Office segments.**

*Question 5: "How are the customers distributed across the countries?"*

```
#Group clients by countries
country_client = df.groupby('Country')['Customer
ID'].nunique().reset_index()
country_client.columns = ['Country', 'Customer ID']
```

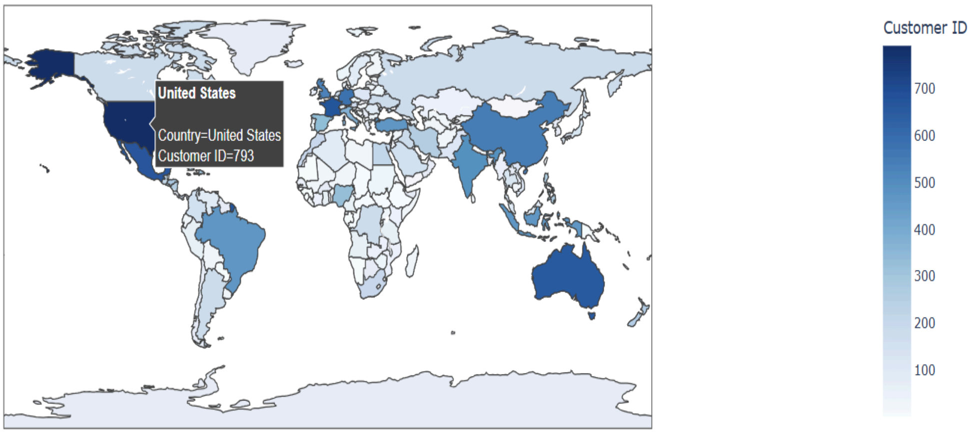
```
#Visualization
import plotly.express as px

fig = px.choropleth(
    country_client,
    locations='Country',
    locationmode='country names',
    color='Customer ID',
```

```
    hover_name='Country',
    color_continuous_scale='Blues',
    title='Customer Distribution across the Countries'
)

fig.show()
```

Customer Distribution across the Countries

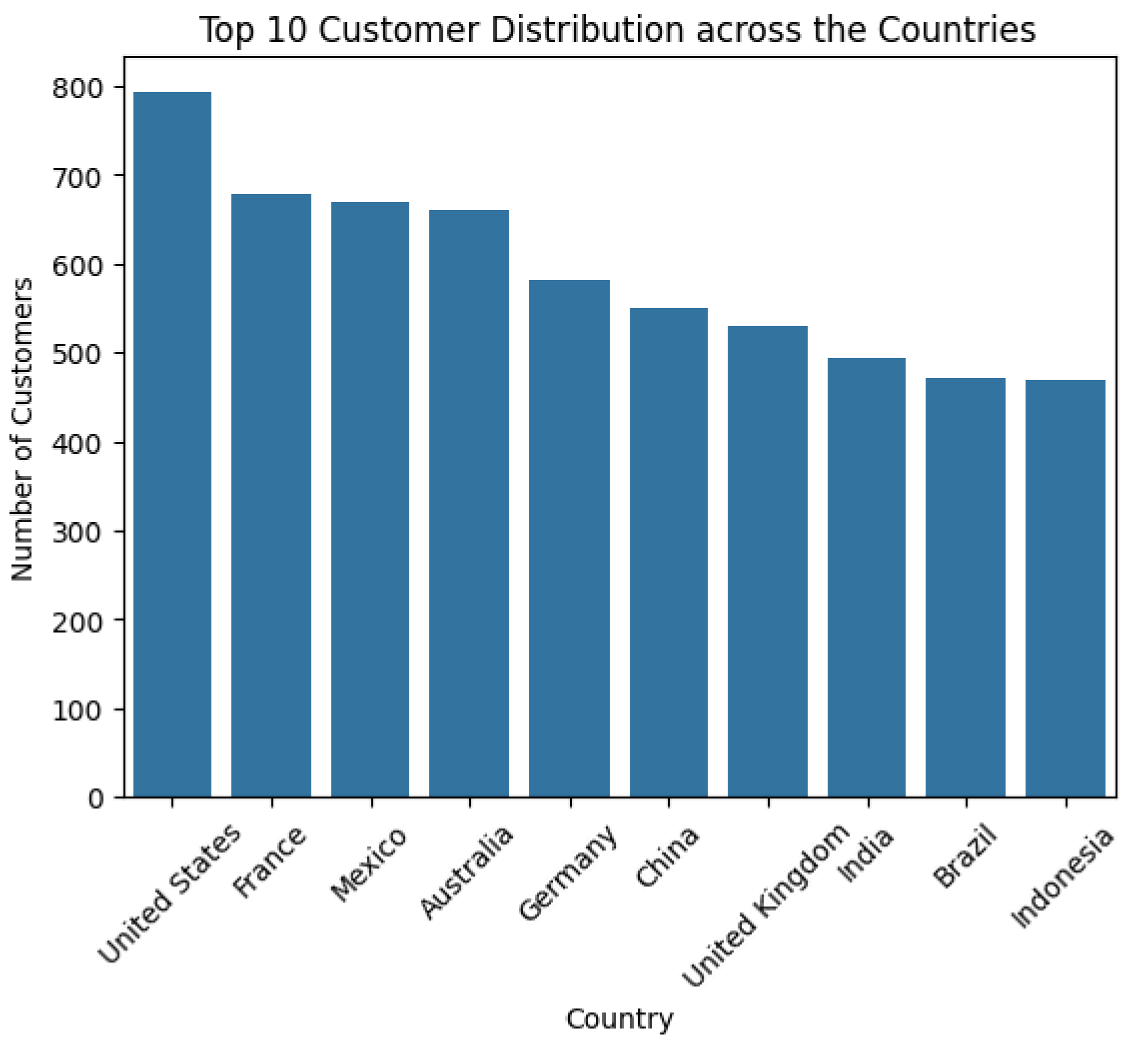



```
#Top 10 countries
country_client_sorted = df.groupby('Country')['Customer ID'].nunique().sort_values(ascending=False)
top_10 = country_client_sorted.head(10)
top_10
```

| | Customer ID |
|---|---|
| **Country** | |
| **United States** | 793 |
| **France** | 679 |
| **Mexico** | 670 |
| **Australia** | 660 |
| **Germany** | 582 |

| | |
|---|---|
| **China** | 549 |
| **United Kingdom** | 529 |
| **India** | 494 |
| **Brazil** | 472 |
| **Indonesia** | 469 |

```
#Visualization
sns.barplot(x= top_10.index, y= top_10.values)
plt.ylabel("Number of Customers")
plt.title('Top 10 Customer Distribution by Country')
plt.xticks(rotation = 45)
plt.show()
```



**The analysis of customer distribution across countries shows a strong concentration of customers in a limited number of regions. The top 10 countries account for the majority of customers, indicating that the business is highly concentrated in specific markets.**

- **Product Analysis**

*Question 1 : "Which country has top sales?"*

```
top_sales =
df.groupby('Country')['Sales'].sum().sort_values(ascending=Fal
se)
top_sales
```

| Country | Sales |
|---|---|
| United States | 2.297201e+06 |
| Australia | 9.252359e+05 |
| France | 8.589311e+05 |
| China | 7.005620e+05 |
| Germany | 6.288400e+05 |
| ... | ... |
| Tajikistan | 2.427840e+02 |
| Macedonia | 2.096400e+02 |
| Eritrea | 1.877400e+02 |
| Armenia | 1.567500e+02 |
| Equatorial Guinea | 1.505100e+02 |

**The United States generates the highest sales, which aligns with its dominant position in terms of customer presence across countries.**

*Question 2 : "Which are the top 5 profit-making product types on a yearly basis?"*

```
#Product profit by year
profit_product = df.groupby(['year', 'Product Name',
'Sub-Category'])['Profit'].sum()
```

```
#Create a new dataframe
profit_product_df = profit_product.reset_index()
```

```
#Sort the dataframe
def top_5_profit_product(group):
  #Sort by descending profit
  group_sorted = group.sort_values(by='Profit', ascending=
False)
  #Keep top 5 profit product
  top_5 = group_sorted.head(5)
  return top_5
```

```
#Top 5 products by year
top_5_products =
profit_product_df.groupby('year').apply(top_5_profit_product)
top_5_products_by_year = top_5_products.reset_index(drop=True)
top_5_products_by_year
```

| | year | Product Name | Sub-Category | Profit |
|---|---|---|---|---|
| **0** | 2011 | Ibico EPK-21 Electric Binding System | Binders | 4630.4755 |
| **1** | 2011 | Cisco Smart Phone, with Caller ID | Phones | 4066.8580 |
| **2** | 2011 | Nokia Smart Phone, Cordless | Phones | 3880.1850 |
| **3** | 2011 | Samsung Smart Phone, VoIP | Phones | 3414.3100 |
| **4** | 2011 | Nokia Smart Phone, Full Size | Phones | 3332.4630 |
| **5** | 2012 | Fellowes PB500 Electric Punch Plastic Comb Bin... | Binders | 7498.8410 |
| **6** | 2012 | Zebra ZM400 Thermal Label Printer | Machines | 3343.5360 |
| **7** | 2012 | Samsung Smart Phone, Full Size | Phones | 2978.8680 |
| **8** | 2012 | Nokia Smart Phone, Full Size | Phones | 2877.1665 |

| | | | | |
|---|---|---|---|---|
| **9** | 2012 | Samsung Smart Phone, with Caller ID | Phones | 2781.8375 |
| **10** | 2013 | Canon imageCLASS 2200 Advanced Copier | Copiers | 9519.9728 |
| **11** | 2013 | Motorola Smart Phone, Full Size | Phones | 7529.4882 |
| **12** | 2013 | Harbour Creations Executive Leather Armchair, ... | Chairs | 6424.5296 |
| **13** | 2013 | Cisco Smart Phone, Full Size | Phones | 6305.4030 |
| **14** | 2013 | Belkin Router, USB | Accessories | 4554.8420 |
| **15** | 2014 | Canon imageCLASS 2200 Advanced Copier | Copiers | 15679.9552 |
| **16** | 2014 | Cisco Smart Phone, Full Size | Phones | 7262.3480 |
| **17** | 2014 | Motorola Smart Phone, Full Size | Phones | 6307.5570 |
| **18** | 2014 | Hoover Stove, Red | Appliances | 5123.2340 |
| **19** | 2014 | Sauder Classic Bookcase, Traditional | Bookcases | 4937.9690 |

```
#Partition the table by year
top_5_products_by_year['year'].unique()

df_2011 =
top_5_products_by_year[top_5_products_by_year['year'] == 2011]
df_2012 =
top_5_products_by_year[top_5_products_by_year['year'] == 2012]
df_2013 =
top_5_products_by_year[top_5_products_by_year['year'] == 2013]
df_2014 =
top_5_products_by_year[top_5_products_by_year['year'] == 2014]
```

```
#Top 5 products in 2011
df_2011
```

| | year | Product Name | Sub-Category | Profit |
|---|---|---|---|---|
| **0** | 2011 | Ibico EPK-21 Electric Binding System | Binders | 4630.4755 |
| **1** | 2011 | Cisco Smart Phone, with Caller ID | Phones | 4066.8580 |
| **2** | 2011 | Nokia Smart Phone, Cordless | Phones | 3880.1850 |

| | | | | |
|---|---|---|---|---|
| 3 | 2011 | Samsung Smart Phone, VoIP | Phones | 3414.3100 |
| 4 | 2011 | Nokia Smart Phone, Full Size | Phones | 3332.4630 |

```
#Top 5 products in 2012
df_2012
```

| | year | Product Name | Sub-Category | Profit |
|---|---|---|---|---|
| 5 | 2012 | Fellowes PB500 Electric Punch Plastic Comb Bin... | Binders | 7498.8410 |
| 6 | 2012 | Zebra ZM400 Thermal Label Printer | Machines | 3343.5360 |
| 7 | 2012 | Samsung Smart Phone, Full Size | Phones | 2978.8680 |
| 8 | 2012 | Nokia Smart Phone, Full Size | Phones | 2877.1665 |
| 9 | 2012 | Samsung Smart Phone, with Caller ID | Phones | 2781.8375 |

```
#Top 5 products in 2013
df_2013
```

| | year | Product Name | Sub-Category | Profit |
|---|---|---|---|---|
| 10 | 2013 | Canon imageCLASS 2200 Advanced Copier | Copiers | 9519.9728 |
| 11 | 2013 | Motorola Smart Phone, Full Size | Phones | 7529.4882 |
| 12 | 2013 | Harbour Creations Executive Leather Armchair, ... | Chairs | 6424.5296 |
| 13 | 2013 | Cisco Smart Phone, Full Size | Phones | 6305.4030 |
| 14 | 2013 | Belkin Router, USB | Accessories | 4554.8420 |

```
#Top 5 products in 2014
df_2014
```

| | year | Product Name | Sub-Category | Profit |
|---|---|---|---|---|
| 15 | 2014 | Canon imageCLASS 2200 Advanced Copier | Copiers | 15679.9552 |
| 16 | 2014 | Cisco Smart Phone, Full Size | Phones | 7262.3480 |

| | | | | |
|---|---|---|---|---|
| **17** | 2014 | Motorola Smart Phone, Full Size | Phones | 6307.5570 |
| **18** | 2014 | Hoover Stove, Red | Appliances | 5123.2340 |
| **19** | 2014 | Sauder Classic Bookcase, Traditional | Bookcases | 4937.9690 |

**Although individual products vary slightly from year to year, the top 5 most profitable products each year are largely dominated by recurring sub-categories, particularly Phones, Binders, and Copiers. This suggests a stable demand pattern across product categories over time.**

*Question 3:* "How is the product price varying with sales? Is there any increase in sales with the decrease in price at a day level?"

```
#Average price per day
df['Price'] = df['Sales'] / df['Quantity']
mean_price = df.groupby(['Product Name', 'Order Date'])['Price'].mean()
```

```
#Quantity sold per day
quantity_sold = df.groupby(['Product Name', 'Order Date'])['Quantity'].sum()
```

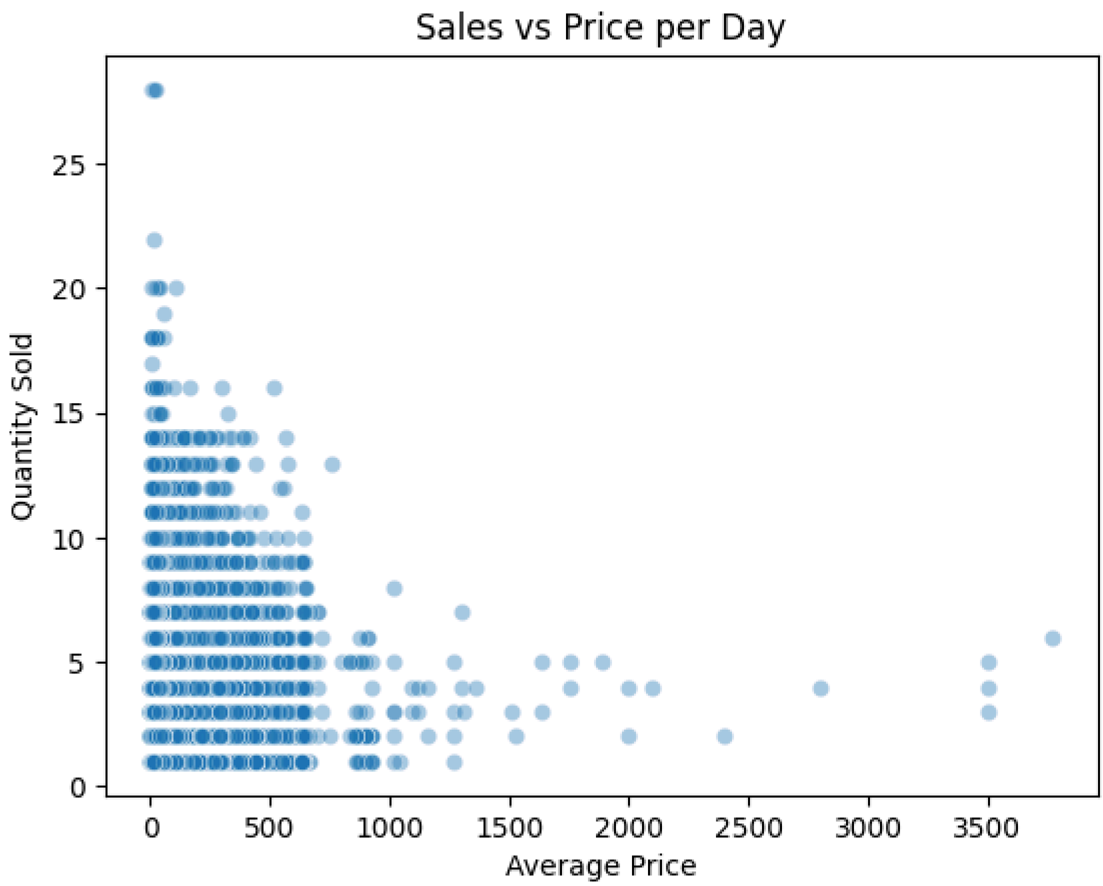
```
daily_data = df.groupby(['Product Name', 'Order Date']).agg({'Price': 'mean', 'Quantity':'sum'}).reset_index()
daily_data
```

| | Product Name | Order Date | Price | Quantity |
|---|---|---|---|---|
| **0** | "While you Were Out" Message Book, One Form pe... | 2014-09-04 | 2.968 | 3 |
| **1** | "While you Were Out" Message Book, One Form pe... | 2014-10-31 | 3.710 | 2 |

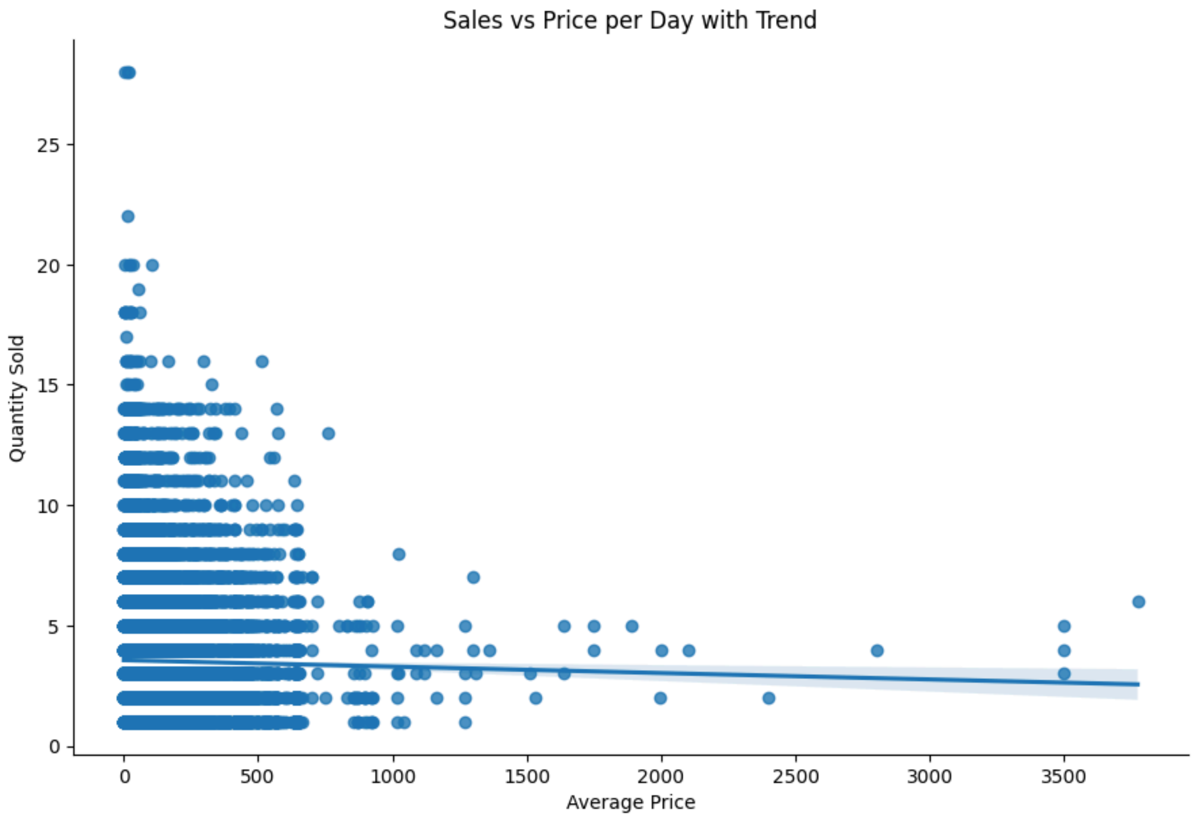
| | | | | |
|---|---|---|---|---|
| **2** | "While you Were Out" Message Book, One Form pe... | 2014-11-14 | 2.968 | 3 |
| **3** | #10 Gummed Flap White Envelopes, 100/Box | 2012-11-03 | 3.304 | 2 |
| **4** | #10 Gummed Flap White Envelopes, 100/Box | 2013-01-15 | 4.130 | 4 |
| **...** | ... | ... | ... | ... |
| **50405** | netTALK DUO VoIP Telephone Service | 2011-12-05 | 41.992 | 4 |
| **50406** | netTALK DUO VoIP Telephone Service | 2012-06-20 | 41.992 | 3 |
| **50407** | netTALK DUO VoIP Telephone Service | 2012-09-25 | 52.490 | 2 |
| **50408** | netTALK DUO VoIP Telephone Service | 2013-01-09 | 41.992 | 9 |
| **50409** | netTALK DUO VoIP Telephone Service | 2014-07-04 | 41.992 | 4 |

50410 rows × 4 columns

```
#Visualization
sns.scatterplot(data= daily_data, x='Price', y='Quantity',
alpha= 0.4)
plt.xlabel("Average Price")
plt.ylabel("Quantity Sold")
plt.title("Sales vs Price per Day")
plt.show()
```




```
#Trend
sns.lmplot(data=daily_data, x='Price', y='Quantity', height=6,
aspect=1.5)
plt.xlabel("Average Price")
plt.ylabel("Quantity Sold")
plt.title("Sales vs Price per Day with Trend")
plt.show()
```




**Sales tend to increase when prices decrease, suggesting an inverse relationship between price and sales at a daily level.**

*Question 4: “What is the average delivery time across the counties?”*

```
#Delivery time
df['Delivery_time_day'] = df['Ship Date'] - df['Order Date']
```

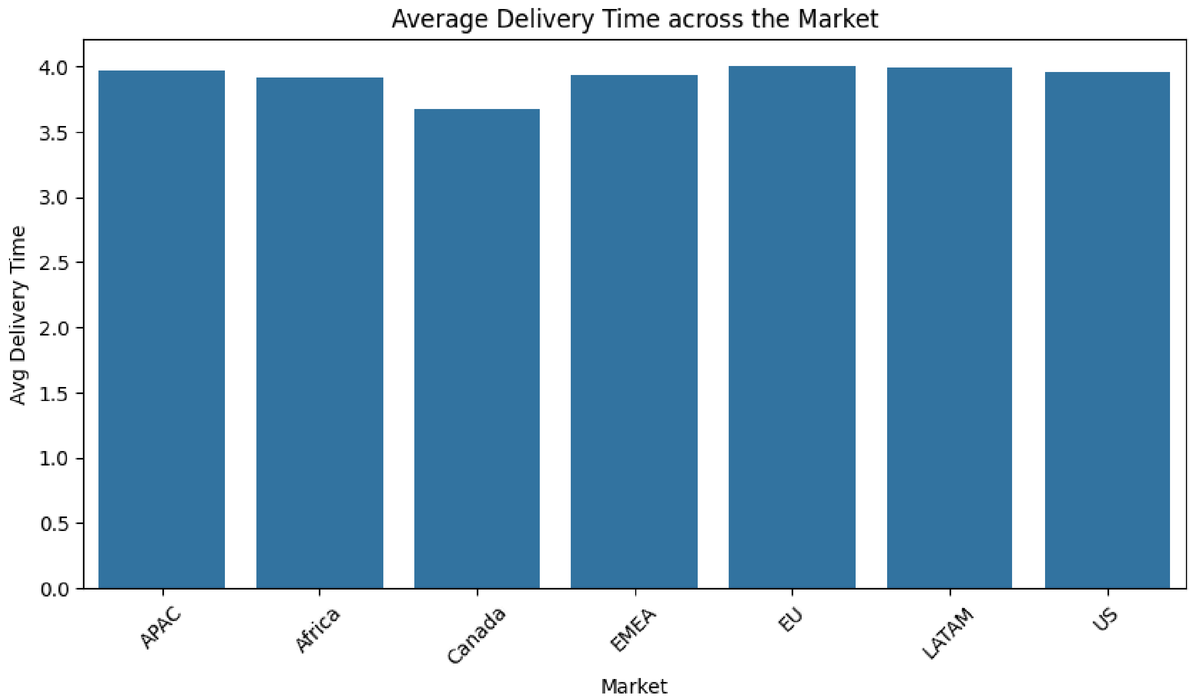
```
df['Delivery_time_days'] = df['Delivery_time_day'] /
pd.Timedelta(days=1)
```

```
#Average delivery time by region
avg_time_market =
df.groupby('Market')['Delivery_time_days'].mean()
avg_time_market
```

| | Delivery_time_days |
|---|---|
| **Market** | |
| **APAC** | 3.969097 |
| **Africa** | 3.910399 |
| **Canada** | 3.677083 |
| **EMEA** | 3.933386 |
| **EU** | 4.008300 |
| **LATAM** | 3.996794 |
| **US** | 3.958875 |

```
avg_time = avg_time_market.reset_index()
```

```
#Barplot
plt.figure(figsize=(10,5))
sns.barplot(data=avg_time, x='Market', y='Delivery_time_days')
plt.xlabel("Market")
plt.ylabel("Avg Delivery Time")
plt.title("Average Delivery Time by Market")
plt.xticks(rotation=45)
plt.show()
```




**Average delivery times are relatively consistent across countries, with only slight variations. Canada shows marginally faster delivery times compared to other markets.**

# KEY INSIGHTS

The analysis shows that high-frequency customers generate the highest revenue and profit, indicating a positive relationship between purchase frequency and profitability. In contrast, a negative relationship is observed between prices and sales, suggesting that customers tend to purchase less when prices increase.

The business activity is concentrated in a limited number of key markets, particularly the United States, France, Mexico, and Australia. The United States also ranks first in terms of total sales across countries.

The analysis further reveals a stable demand for certain product sub-categories over time, particularly Phones, Binders, and Copiers, which consistently appear among the top-performing segments each year.

Finally, average delivery times are relatively consistent across global markets, indicating comparable levels of operational performance across countries.

# CONCLUSION

This analysis explored customer behavior and product performance using the Global Superstore dataset, with the aim of identifying key drivers of revenue, profitability, and operational efficiency.

The results show that high-frequency customers are the main contributors to both revenue and profit, while pricing has an inverse relationship with sales at a daily level. The business activity is concentrated in a few key markets, particularly the United States, which also leads in total sales. In addition, a limited number of product sub-categories consistently dominate profitability over time, indicating stable demand patterns. Finally, delivery times remain relatively consistent across countries, suggesting a uniform level of operational performance.

Overall, these findings highlight the importance of focusing on high-value customers and key product categories, while also optimizing pricing strategies and maintaining strong performance in core markets.